WEB TECHNOLOGIES LAB – IT 5512

MINI PROJECT

PUBLISHER-SUBSCRIBER NETWORK

PROJECT REPORT

Submitted by

Guhan M (2022506065)

Harini R (2022506033)

Mahalakshmi S (2022506024)

MADRAS INSTITUTE OF TECHNOLOGY

ANNA UNIVERSITY, CHENNAI – 600044

INTRODUCTION:

This program implements a Publisher-Subscriber model with a BrokerServer using Java sockets for TCP communication. It features GUIs for the publisher and subscriber, ensuring reliable message delivery.

PROJECT OBJECTIVE

Project Goal:

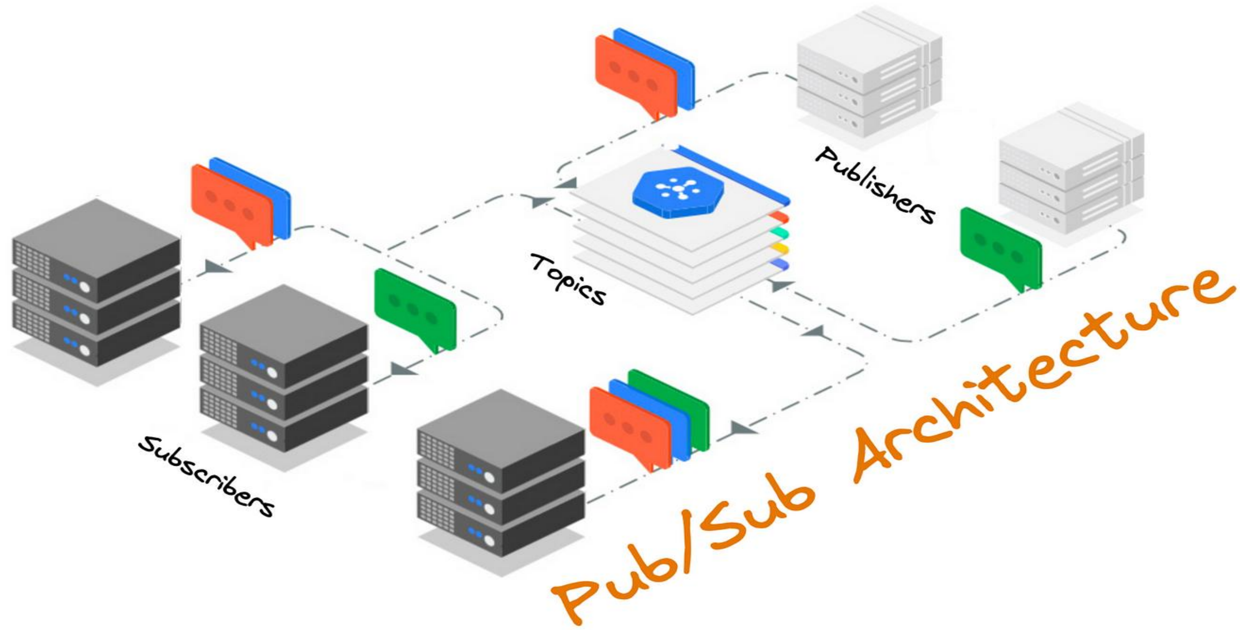
Develop a simple Publisher-Subscriber system using Java sockets, where:

- **Publisher:** Sends messages to specific topics and can create new topics.
- **Subscriber:** Receives messages from subscribed topics and can dynamically manage subscriptions.
- **BrokerServer:** Forwards messages from publishers to topic subscribers, handles subscriptions, and manages connections.

System Components:

1. **Publisher:**
    - Creates topics and publishes messages.
2. **BrokerServer:**
    - Manages connections for publishers and subscribers.
    - Forwards messages to topic subscribers.
    - Handles "subscribe" and "unsubscribe" commands.
3. **Subscriber:**
    - Subscribes to topics and receives messages dynamically.
4. **GUI:**
    - Simplifies interactions with publishers and subscribers.
    - Features options for creating topics, subscribing, and publishing via dropdown menus.

ARCHITECTURE:



TECH STACK USED:

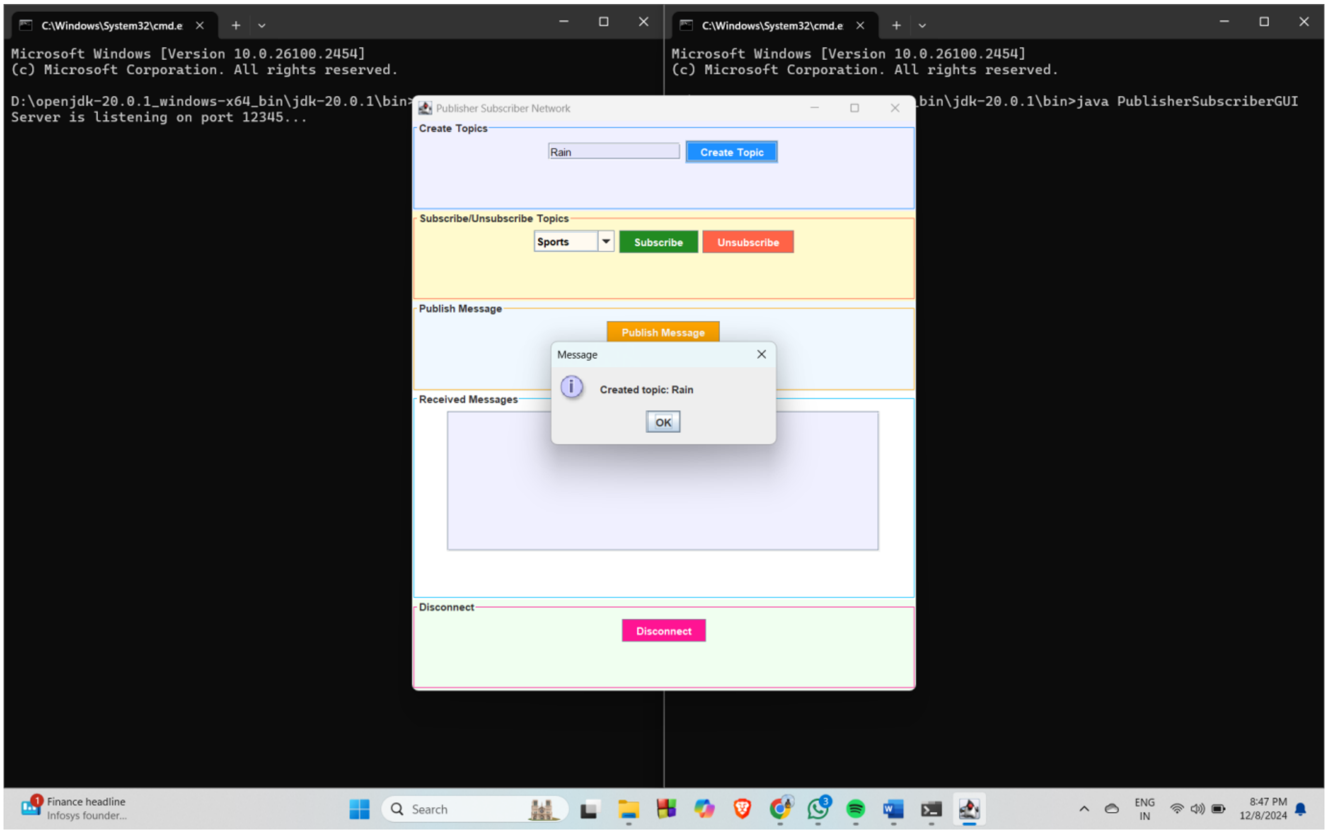
Core Technologies:

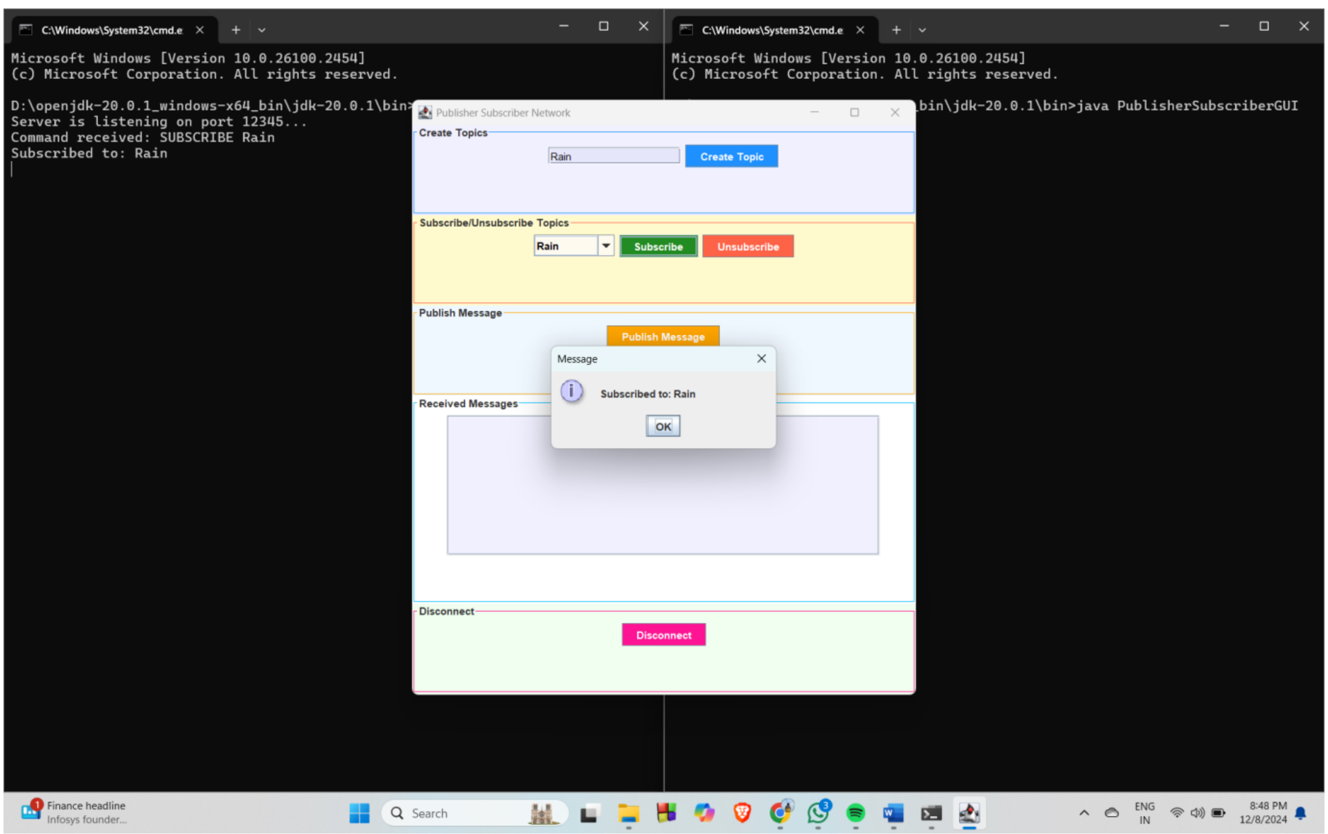
1. **Programming Language:**
   - **Java**: Used for implementing the Publisher, Subscriber, and BrokerServer logic.
2. **Networking:**
   - **Java Sockets (TCP/IP):** Facilitates communication between the Publisher, Subscriber, and BrokerServer, ensuring reliable data transfer.
3. **GUI Framework:**
   - **Java Swing:** Provides graphical user interfaces for the Publisher and Subscriber.

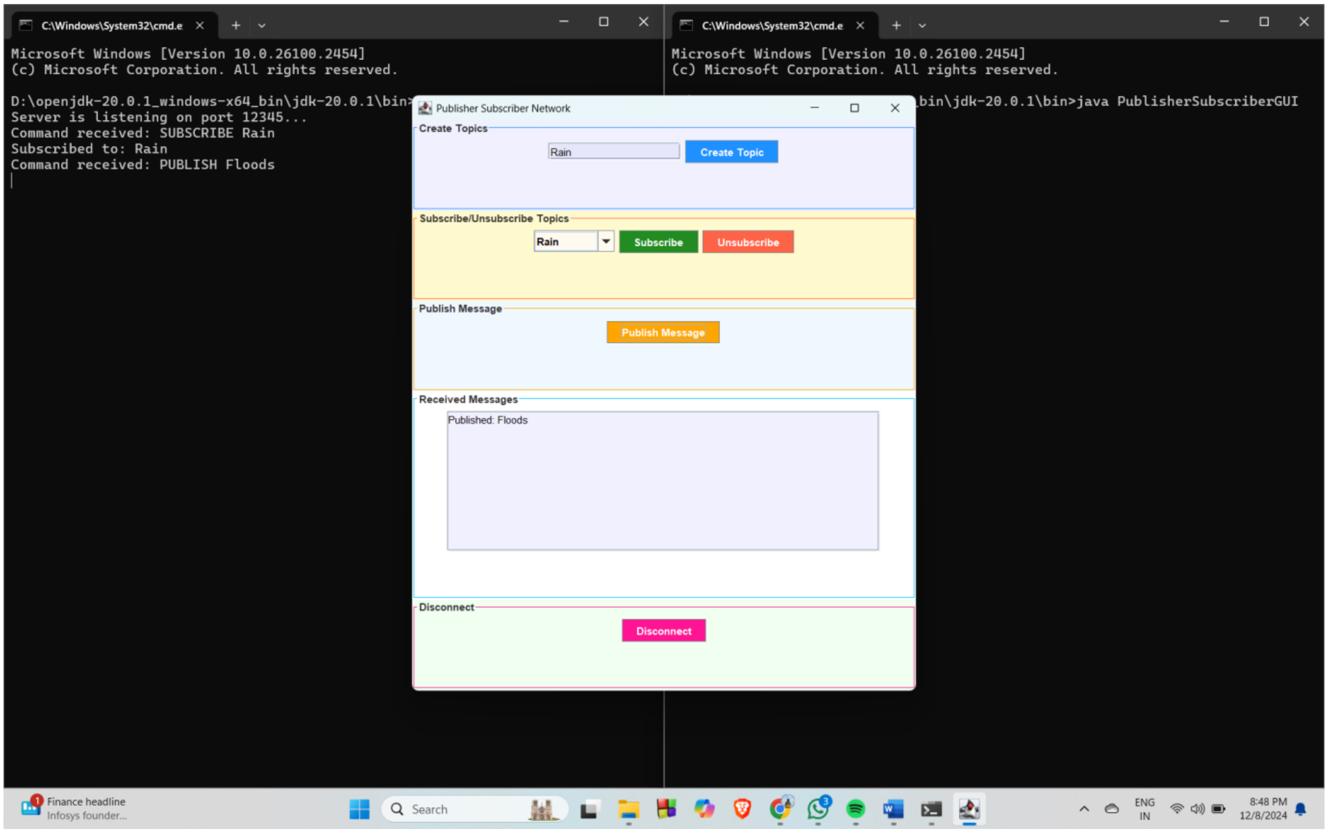
Libraries and Utilities:

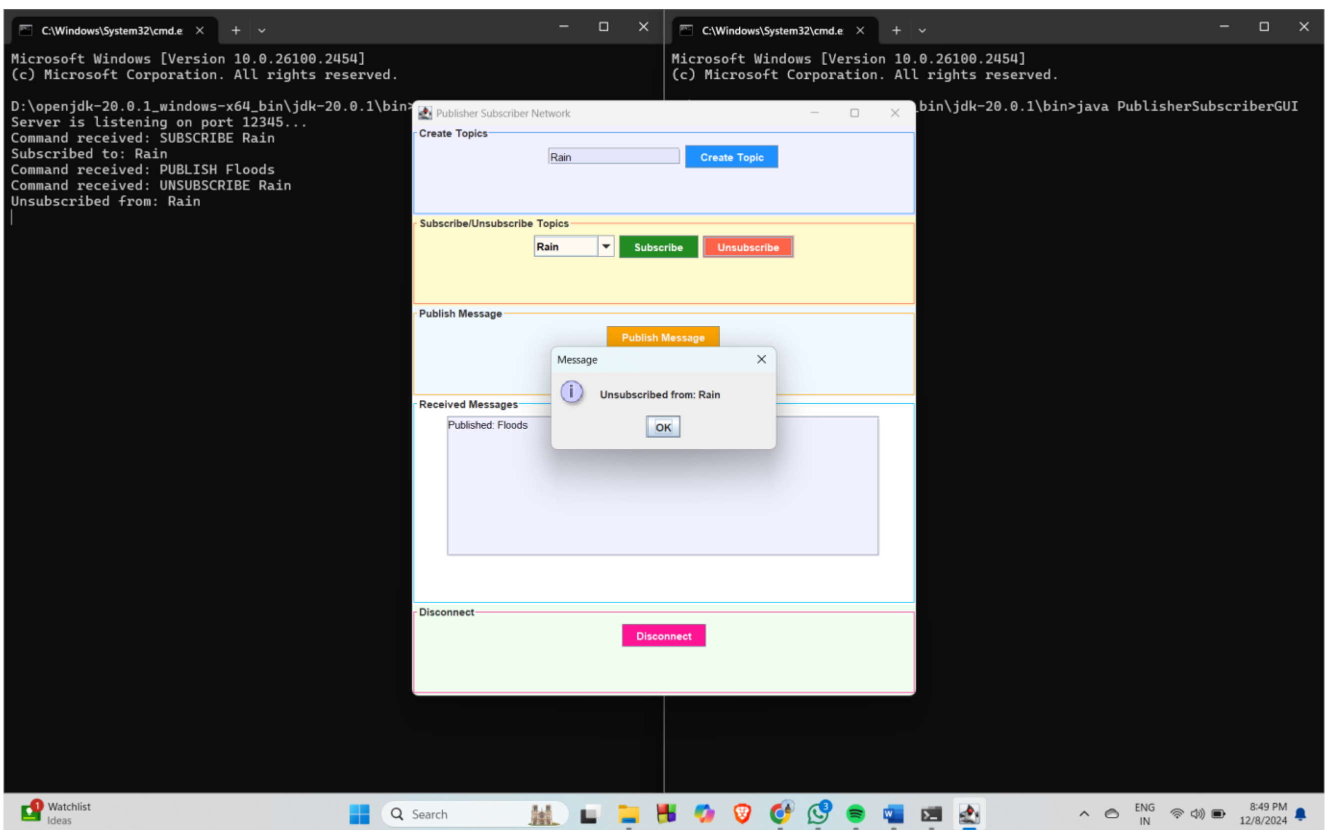
1. **I/O Streams:**
   - **BufferedReader** and **PrintWriter**: Handle input/output streams for socket communication.
2. **Threading:**
   - **Java Threads:** Enable concurrent handling of multiple clients (Publishers and Subscribers) in the BrokerServer.
3. **Data Structures:**
   - **HashMap/List:** Used for managing topics and subscriptions in the BrokerServer.

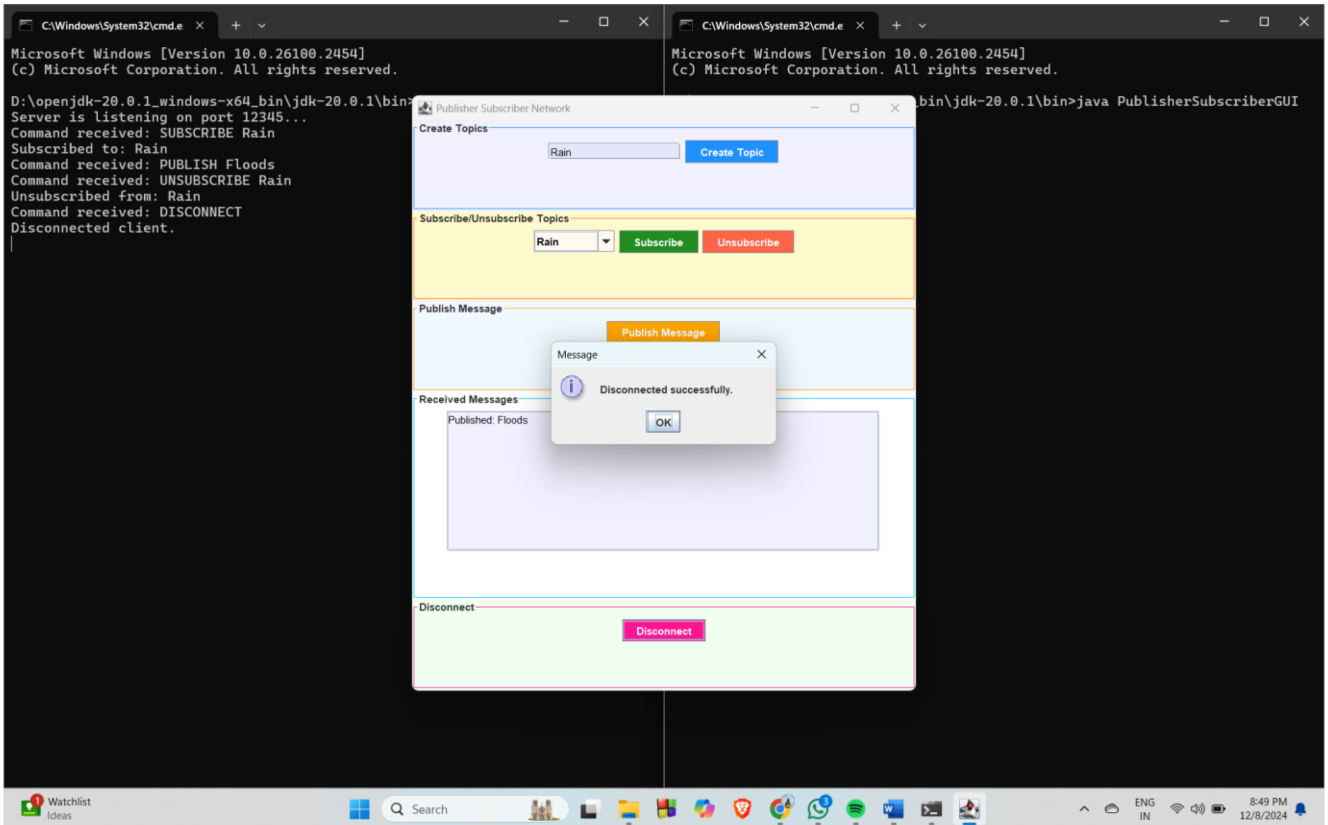
OUTPUT SCREENSHOTS:






C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26100.2454]
(c) Microsoft Corporation. All rights reserved.
D:\openjdk-20.0.1_windows-x64_bin\jdk-20.0.1\bin
Server is listening on port 12345...
Command received: SUBSCRIBE Rain
Subscribed to: Rain
Command received: PUBLISH Floods
bin\jdk-20.0.1\bin>java PublisherSubscriberGUI
Publisher Subscriber Network
Create Topics
Rain
Create Topic
Subscribe/Unsubscribe Topics
Rain
Subscribe
Unsubscribe
Publish Message
Publish Message
Received Messages
Published: Floods
Disconnect
Disconnect
Finance headline
Infosys founder...
Search
ENG
IN
8:48 PM
12/8/2024


C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26100.2454]
(c) Microsoft Corporation. All rights reserved.
D:\openjdk-20.0.1_windows-x64_bin\jdk-20.0.1\bin
Server is listening on port 12345...
Command received: SUBSCRIBE Rain
Subscribed to: Rain
Command received: PUBLISH Floods
Command received: UNSUBSCRIBE Rain
Unsubscribed from: Rain
bin\jdk-20.0.1\bin>java PublisherSubscriberGUI
Publisher Subscriber Network
Create Topics
Rain
Create Topic
Subscribe/Unsubscribe Topics
Rain
Subscribe
Unsubscribe
Publish Message
Publish Message
Message
Unsubscribed from: Rain
OK
Received Messages
Published: Floods
Disconnect
Disconnect
Watchlist
Ideas
Search
ENG
IN
8:49 PM
12/8/2024



CONCLUSION:

This program implements a robust Publisher-Subscriber system using Java sockets and a GUI. The broker manages message broadcasting and subscriptions, while the GUI enables seamless user interactions for creating topics, publishing messages, and managing.